



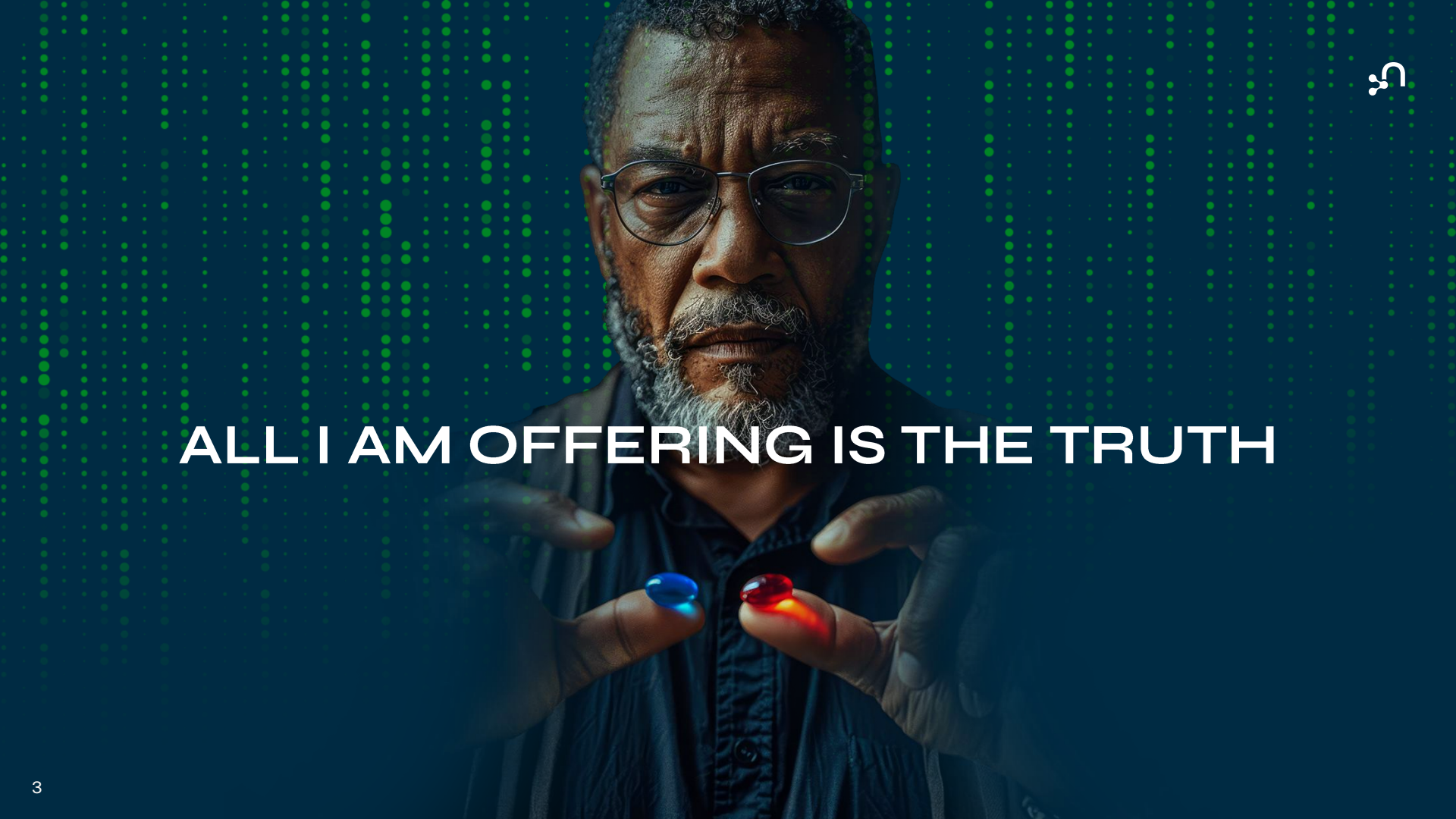
Connecting the dots with context graphs

Stephen Chin (@steveonjava)

VP of Developer Relations @ Neo4j



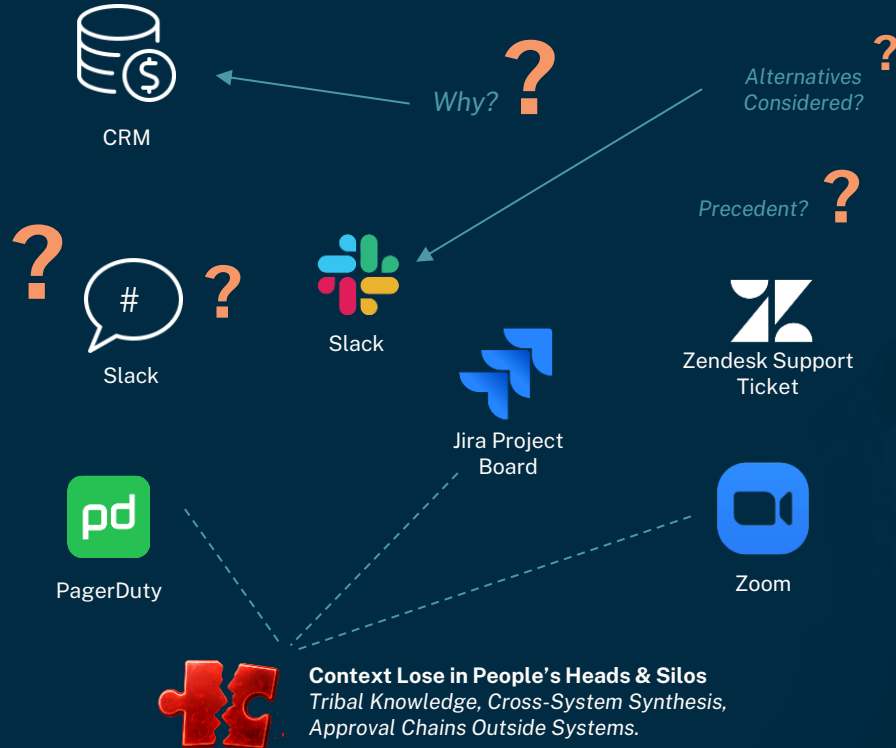




ALL I AM OFFERING IS THE TRUTH

Today

Scattered, lost reasoning



Context Graph

Connected, captured decision traces

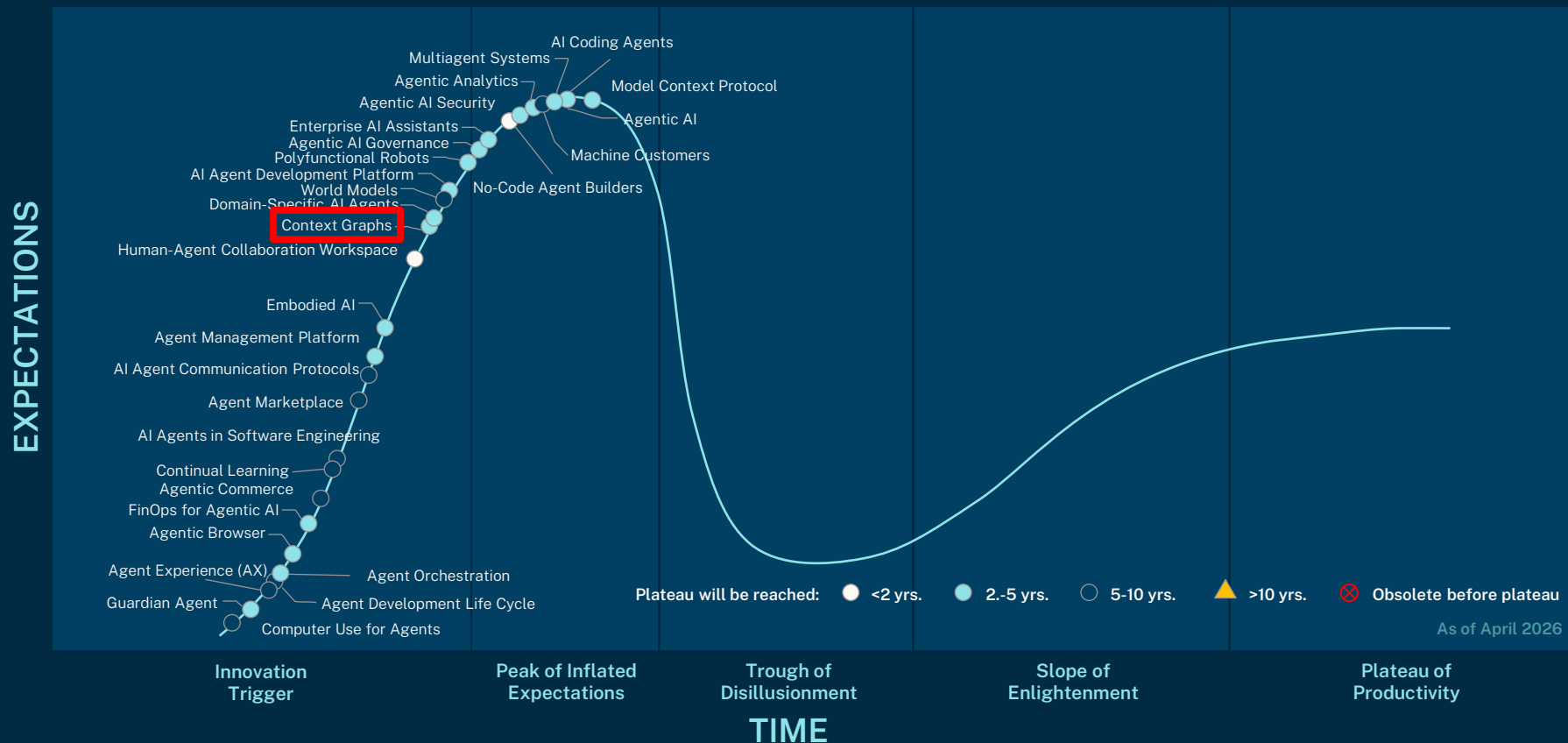


*Inputs, Policies, Exceptions, Precedents,
Approvers, Approvers, Outcomes.*



**Decision Trace as First-
Class Data**

Hype cycle for Agentic AI







Knowledge Graphs



Knowledge graph components



Nodes

represent entities in the graph

Relationships

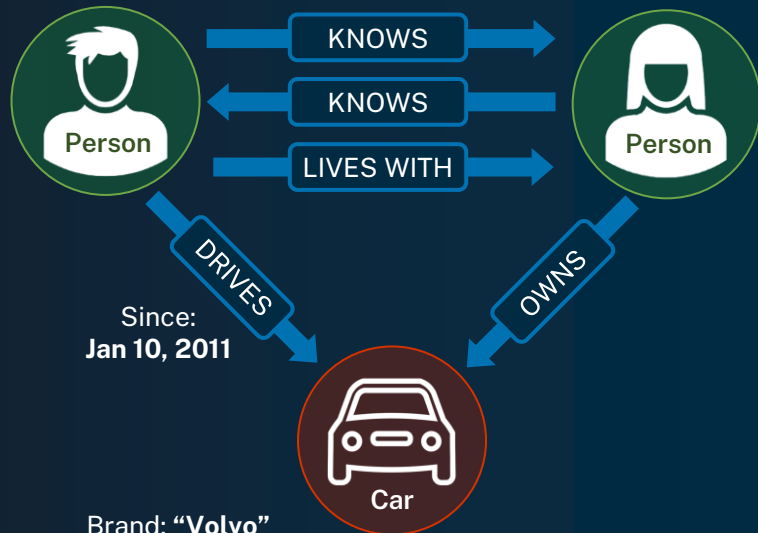
represent associations or interactions between nodes

Properties

represent attributes of nodes or relationships including vectors, can be indexed

Name: "Dan"
Born: May 29, 1970
Twitter: "@dan"

Name: "Ann"
Born: Dec 5, 1975

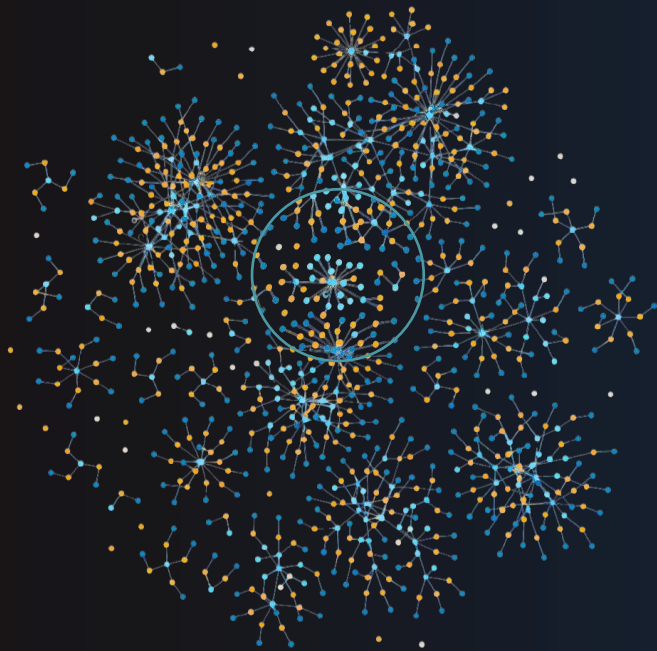


Since:
Jan 10, 2011

Brand: "Volvo"
Model: "V70"
Description: "An executive car manufactured and..."
DescEmbedding: [0.1, -0.3, 0.4, ..., -0.7]
DescSource: "https://en.wikipedia.org/wiki/Volvo_V70"

Context from graphs enables explainable AI

How do you ensure a high-quality production environment with Agents?



Storing

learnings from user/agent interactions in the graph database as context



Visualizing

conversations, flows, reasoning within context



Analysing

Context data for agent system performance, semi-automatically identify & apply opportunities for improvement

LLMs and Knowledge Graphs



BETTER TOGETHER

Enhance relevance with domain context



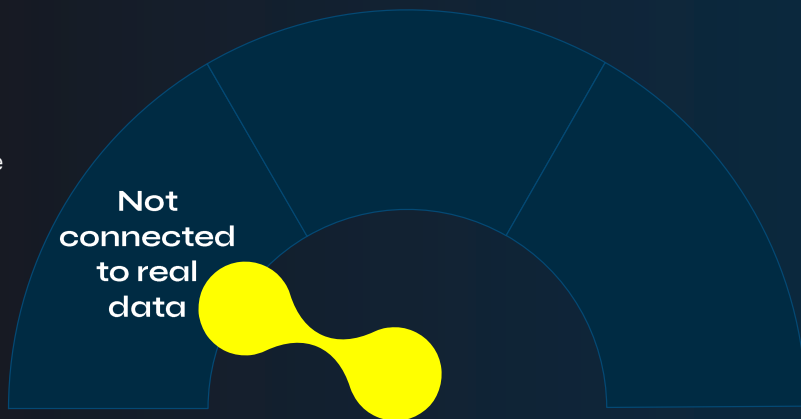
What was in the Care Plans associated with Andrea Jenkins's emphysema?

LLM Direct

OpenAI function call

The care plan associated with Andrea7 Jenkins714 emphysema likely included the following components:

Preventing further damage to the lungs:
This may involve advising the patient to...



Enhance relevance with domain context



What was in the Care Plans associated with Andrea Jenkins's emphysema?

Baseline RAG

**LlamaIndex-RAG backed by GPT 4.0
and Chroma vector database**

The Care Plan associated with Andrea Jenkins' emphysema included activities **respiratory therapy, specifically deep breathing and coughing exercises.**

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OpenAI over Neo4j function call

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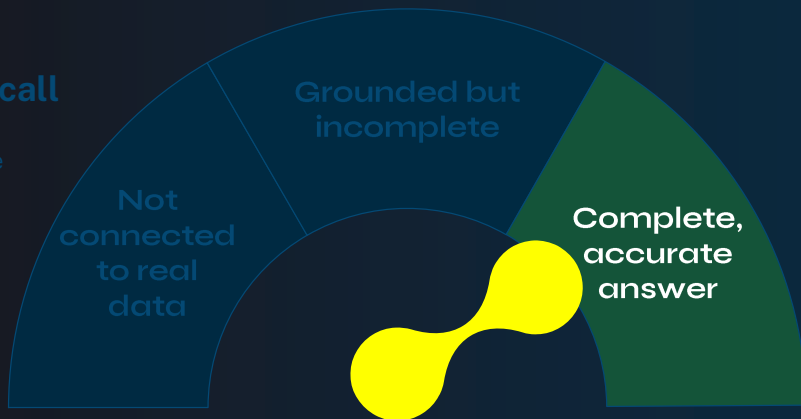
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GraphRAG

Neo4j, LangChain, and OpenAI

The Care Plan for Andrea7 Jenkins714's emphysema included **medication management, smoking cessation counseling, and pulmonary rehabilitation exercises.**

A portrait of actor Nico Kidman with long, wavy brown hair, a beard, and sunglasses. He is wearing a blue martial arts gi. The background is dark blue with a pattern of green dots and faint, illegible text. The text "I KNOW KUNG FU" is overlaid in white, bold, sans-serif font across the center of the image.

I KNOW KUNG FU



Short term memory

- Current Context
- Compression, Relevancy
- Integrate Tool Results

Long term memory

- Episodic
- Semantic / Structural
- Procedural / Instructional

Reasoning memory

- Tool Call Traces
- Reasoning Traces
- Previous Decisions

Short-term memory



Conversation history and session state with automatic entity extraction

Conversation storage

Sessions and messages persisted as graph nodes with metadata

Multi-stage entity extraction

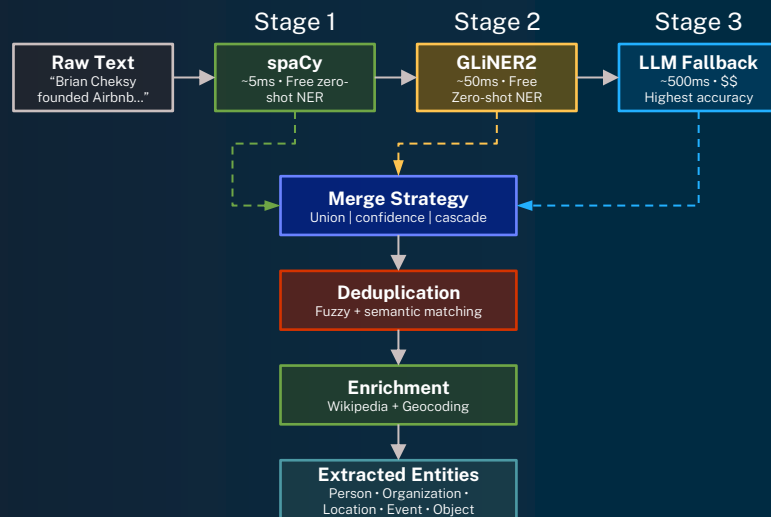
Pipeline combining spaCy, GLiNER, and LLM extractors with configurable merge strategies

Entity resolution

Multi-strategy dedup: exact, fuzzy, and semantic matching with type-aware resolution

```
session = await memory.add_session("user_123")
await memory.add_message(session.id, role="user",
content="Review Jessica's account")
```

Neo4j agent memory entity extraction pipeline



Long-term memory



Persistent knowledge graph of entities, relationships, and learned preferences

POLE+O Entity Model

	Person	Customers, employees, stakeholders
	Organization	Companies, teams, departments
	Location	Addresses, regions, countries
	Event	Transactions, decisions, meetings
	Object	Accounts, products, documents

Key Capabilities

-  Cross-conversation knowledge persistence
-  Temporal relationships (valid from/to)
-  Configurable entity subtypes
-  Automatic extraction from conversations
-  Vector search across all entity types

Reasoning memory



Decision traces, tool usage audits, and provenance — the layer that makes AI explainable

Tool call traces

Every tool invocation, parameters, results — complete audit trail

1

2

Decision provenance

Why did the agent choose this path?
Causal chain fully recorded

Learning from experience

Agent checks if it solved something similar before, reuses successful patterns

3

4

Compliance & debugging

When something goes wrong, you can trace exactly what happened and why



STATUS:
ONLINE

OPERATOR, WE NEED AN EXIT

Why graphs for agent memory?



Relationships Are First-Class

Connections between entities, decisions, and events are the data — not an afterthought



Structural Similarity

Graph embeddings (FastRP) find similar situations by network topology, not just text



Cross-Session Knowledge

Information learned in one conversation is available in all future interactions



Multi-Hop Traversal

Follow chains: Customer → Account → Transaction → Decision → Policy → Employee



Explainable Decisions

Full provenance chain: trace exactly how and why every decision was made



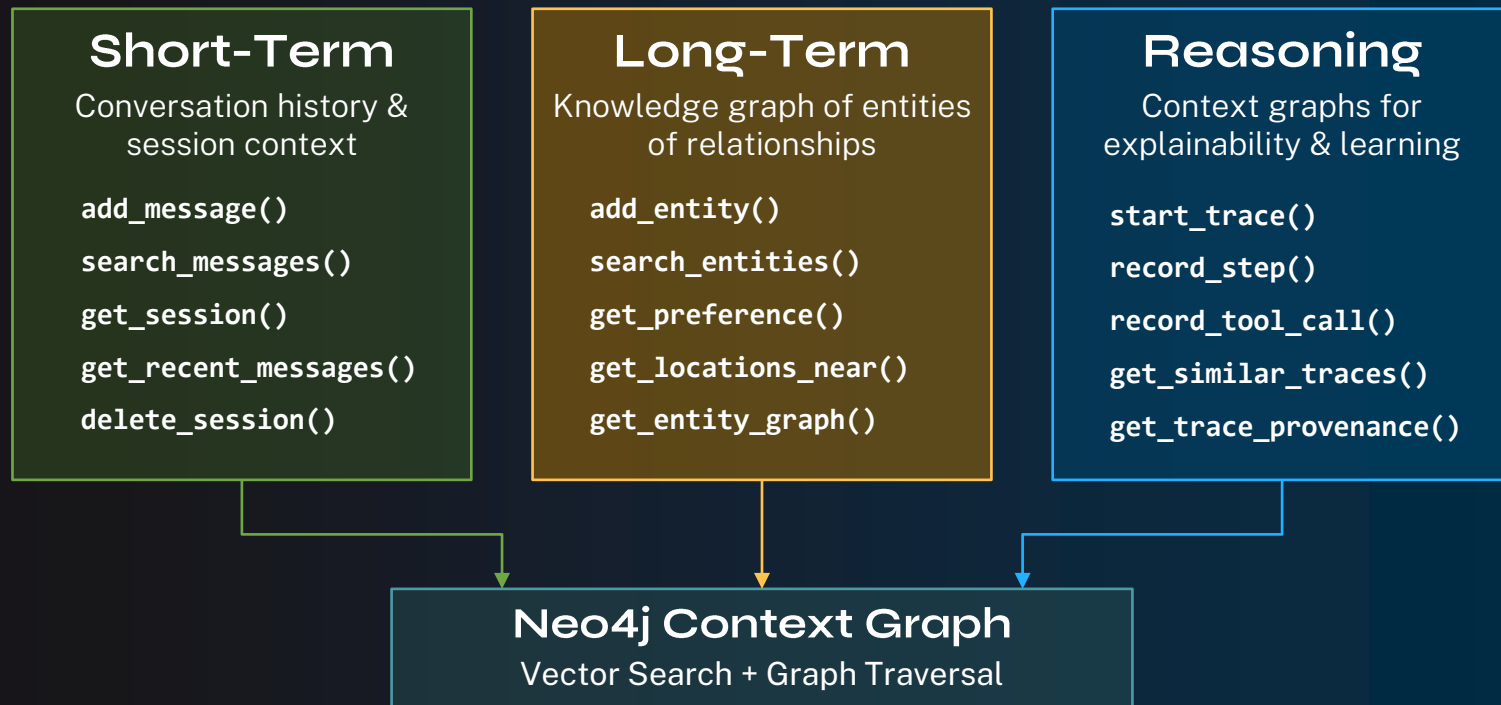
Production Ready

Neo4j: ACID compliance, enterprise scale, proven technology

Neo4j-agent memory



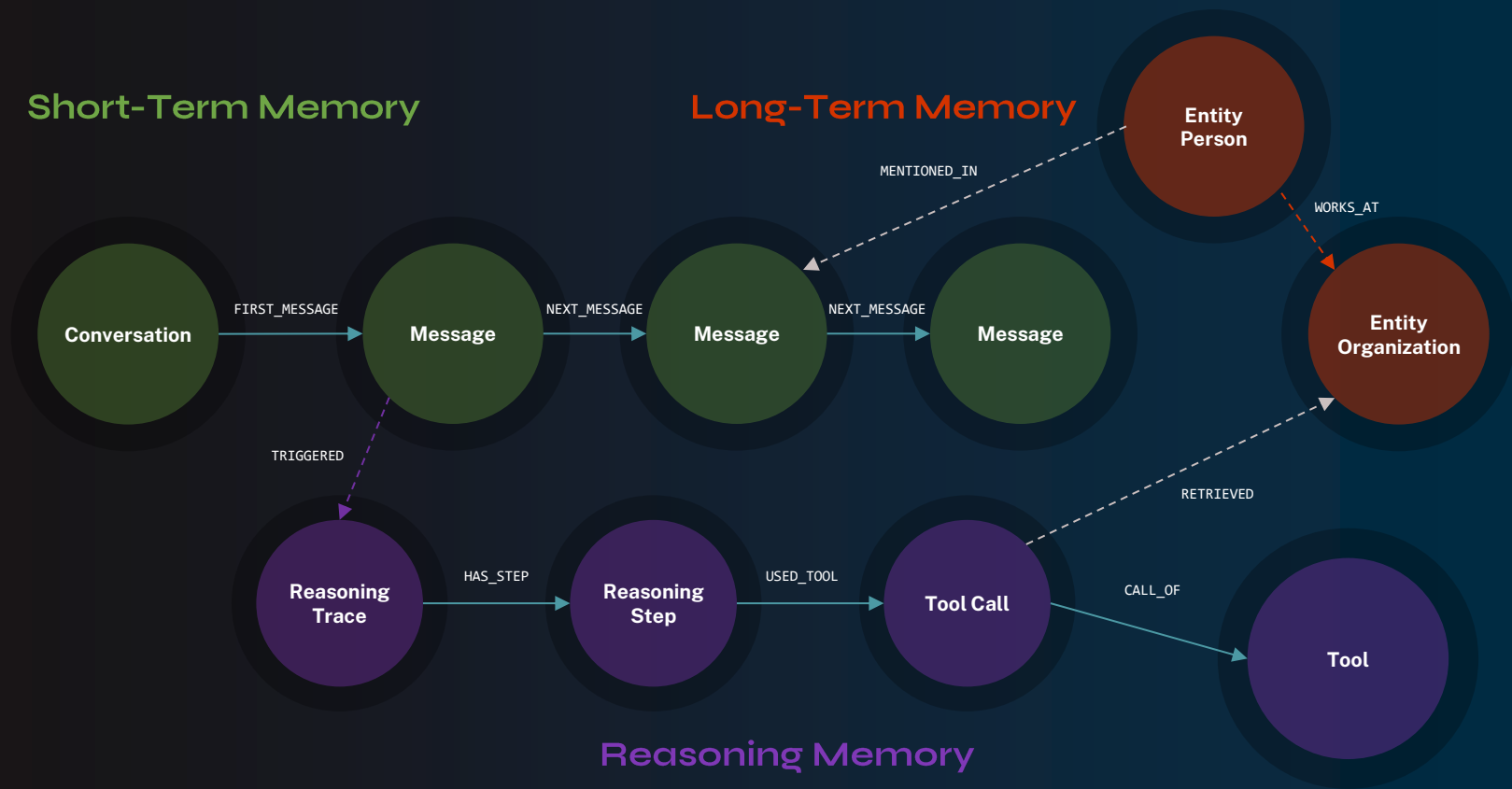
Complete Agent Memory API





Short-Term Memory

Long-Term Memory



Lenny's memory architecture



What the Agent Does

✓ Knowledge Graph Search

Traverse entities (people, companies, topics) and their relationships across episodes

✓ Conversation Memory

Remembers your history and learned preferences, tailors recommendations

✓ Reasoning Reuse

Checks if it solved something similar before, reuses successful patterns

✓ Interactive Visualization

Explore how guests, companies, and topics connect — tool call result cards



Lenny's memory demo



Click a topic to start a new conversation

4/8/2026

4/8/2026

4/8/2026

Use memory graph search to

Powered by

 @neo4j-labs/agent-memory

Neo4j

Lenny's Podcast

 [Blog](#)

Source



Explore insights from 299 podcast episodes with AI-powered graph memory. Click a topic or type your own question.

Use memory graph search to explore what guests say about scaling engineerin...

Tell me about Brian Chesky

What are the most mentioned companies in the podcast?

What entities are related to Airbnb?

What did Brian Chesky say about micromanagement and being in the details?

Show me locations mentioned in the Brian Chesky episode

Ask about Lenny's Podcast...

Press Enter to send, Shift+Enter for new line

Powered by [neo4j-agent-memory](#) | A Neo4j Labs project

[GitHub](#) [Community](#) [Docs](#)

⚙️ Agent Capabilities

Multi-step Reasoning
Plans and executes complex queries step by step

Conversation Memory
Maintains context across messages in the thread

Preference Learning
Adapts responses based on your stored preferences

Knowledge Graph

Queries entities and relationships in Neo4j

Available Tools 21

Tool Call Cards 7

A person in a dark trench coat is walking away from the viewer down a corridor. The corridor is flanked by server racks on both sides, which are illuminated with green light. The background is a dense, vertical rain of green digital characters, creating a 'Matrix' effect. The floor is dark and reflective, showing the green light from the servers. The overall atmosphere is futuristic and high-tech.

WE NEED GRAPHS... LOTS OF GRAPHS

What is a context graph?



A knowledge graph specifically designed to capture decision traces — the full context, reasoning, and causal relationships behind every significant decision.

Traditional Audit Log

Records actions only

"Transaction rejected at 14:32"

No relationships between events

No causal chain or reasoning

Flat, disconnected records

Context Graph

Captures the full "why"

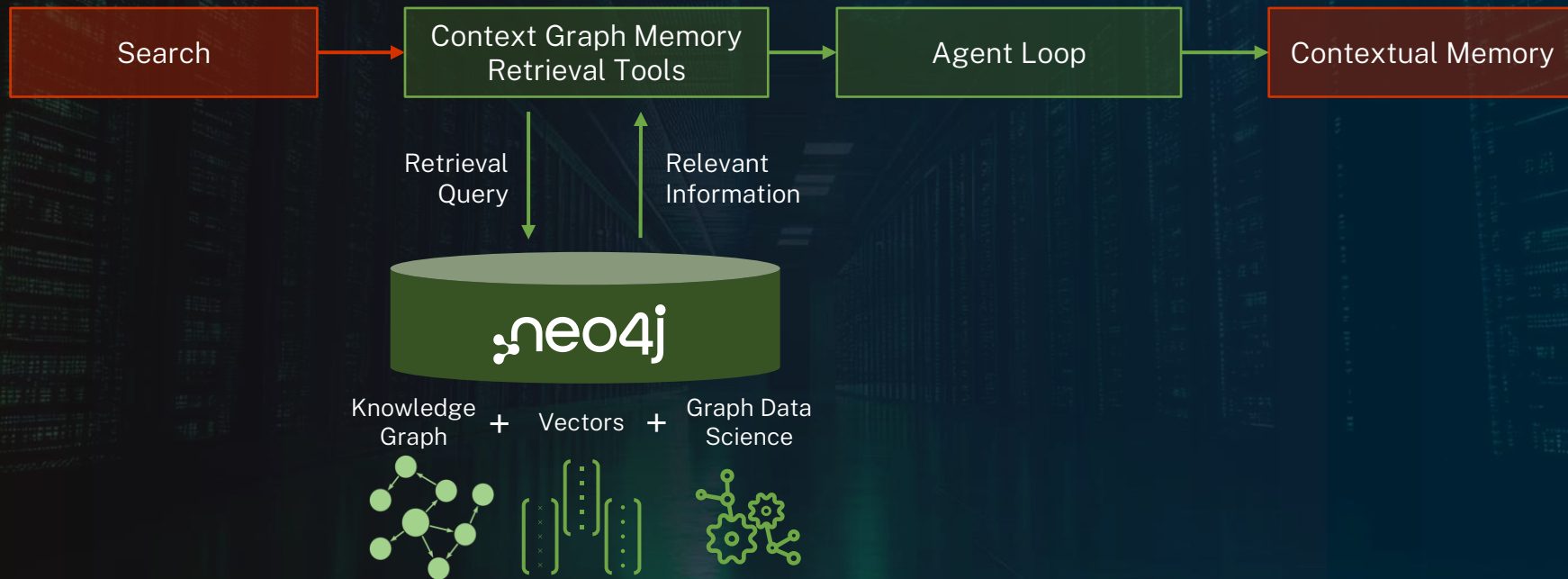
Decision traces & causal chains

Entities, relationships, events

Tribal knowledge made queryable

Connected, traversable structure

Context graph based retrieval



Context graph: financial services data model



Financial Services Context Graph

ENTITIES (What Exists)

Person, Account, Transaction, Organization

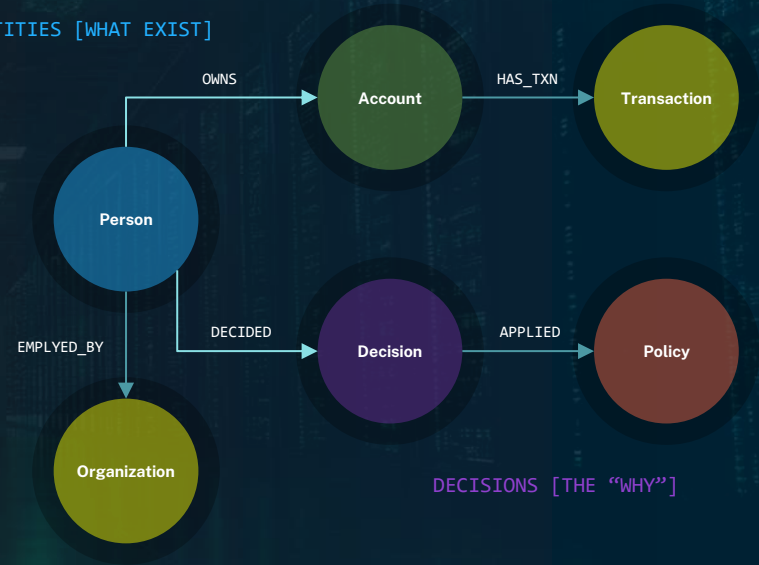
EVENTS (What Happened)

Decisions, Transactions, Approvals, Rejections

CONTEXT (The Why)

Policies Applied, Risk Factors, Employee Reasoning

ENTITIES [WHAT EXIST]

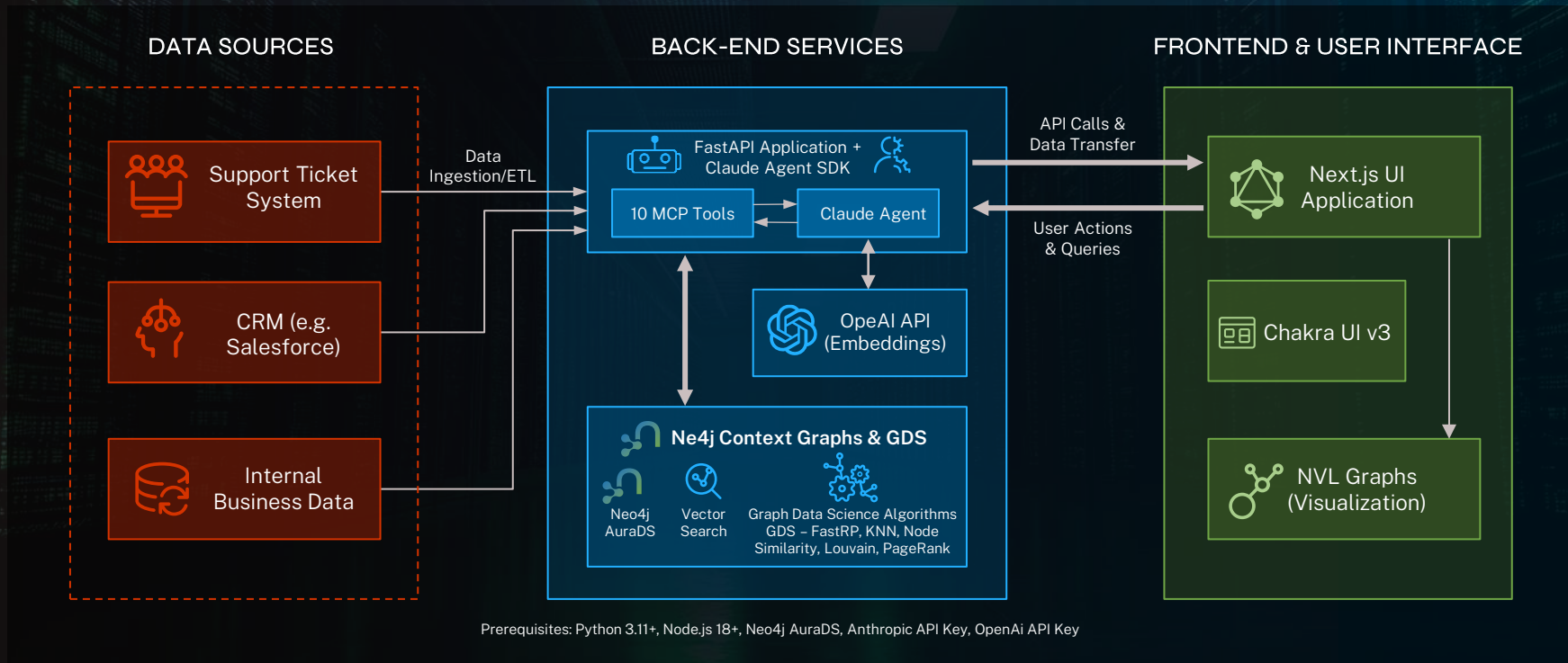


DECISIONS [THE "WHY"]

Blog: neo4j.com/blog/agentic-ai/hands-on-with-context-graphs-and-neo4j/

Context graph demo architecture

Ai-powered decision tracing for financial institutions



Context graph demo





Context Graph Demo

AI-powered decision tracing with Neo4j

[Blog Post](#)

[Issues](#)

[GitHub](#)

[About & Schema](#)

AI Assistant

Ask questions about customers, decisions, and policies

Welcome to Context Graph Demo

I can help you search for customers, analyze decisions, find similar precedents, and trace causal relationships. Try asking:

- Should we approve a credit limit increase for Jessica Norris? She's requesting a \$25,000 limit increase.
- Search for customer John Walsh
- A customer wants to make a \$15,000 wire transfer. What policies apply and are there similar past decisions?
- We need to override the trading limit for Katherine Miller. Find precedents for similar exceptions.

Ask about customers, decisions, or policies...



Context Graph

Visualize entities, decisions, and causal relationships

No graph data to display.

Use the AI assistant to search for customers or decisions to visualize the context graph.





neo4j Graph Academy

Learn everything you need to know to combine Generative AI and knowledge graphs to produce highly accurate responses, with rich context and deep explainability.



<https://dev.neo4j.com/ga-rag>

Tomorrow's AI Unicorns Build on Neo4j



AI is the most ambitious startup frontier in decades. To succeed in this environment, the next unicorn founders need more than ambition. They need speed, reliability, and architecture that adapts as fast as AI evolves. The Neo4j Startup Program equips founders with everything they need to build explainable, scalable, production-ready AI applications.

Founders get support across three core areas:

Runway: Receive credits for Neo4j Aura, a graph-native database built for AI.

Expertise: Validate your GenAI architecture with Neo4j graph engineers.

Exposure: Co-marketing and partnership opportunities to drive visibility and product adoption.

From MVP to IPO, Neo4j is your trusted AI innovation partner — every build, every launch, every win.





Thank you

Stephen Chin (@steveonjava)

VP of Developer Relations @ Neo4j

